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INFORMATION PROCESSING DEVICE

Abstract

The processing device of the information processing device includes: a first step of calculating a relative frequency distribution of the original data; a second step of setting a plurality of time windows for clipping data of a part of the period of the original data; a third step of clipping data from the original data; a fourth step of calculating a relative frequency distribution in the extracted data; and a fifth step of calculating an error between the relative frequency distribution in the original data and the relative frequency distribution in the extracted data; and a search process of repeatedly performing the trial from the second step to the fifth step by changing the setting of the plurality of time windows, and calculating an index value of damage to the engine-disconnecting clutch using the extracted data whose error is equal to or less than the threshold value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-033046 filed on Mar. 5, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to information processing devices.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2008-108247 (JP 2008-108247 A) discloses an information processing device that reduces the size of data for analysis by compressing original data for analysis. The original data for analysis is data collected over a predetermined period using sensors mounted on a vehicle.

[0004] The information processing device disclosed in JP 2008-108247 A compresses data by extracting, from the original data, data acquired at the time when a certain vehicle speed is reached and data acquired at the time of an inflection point of the vehicle speed.

SUMMARY

[0005] The above information processing device extracts data by focusing only on the vehicle speed. Therefore, the above information processing device cannot extract data according to characteristics of data other than the vehicle speed. There is a demand for an information processing device that can obtain extracted data capturing characteristics of the entire original data including features other than the vehicle speed.

[0006] An information processing device for solving the above issue is an information processing device that acquires original data collected and created over a predetermined period using a plurality of sensors mounted on a vehicle and calculates an index value indicating a magnitude of damage accumulated in an engine-disconnecting clutch.

[0007] The information processing device includes a processing device configured to perform a process.

[0008] The original data includes, as features, the number of engagements of the engine-disconnecting clutch and a relative rotational speed between an input shaft and an output shaft of the engine-disconnecting clutch when the engine-disconnecting clutch is engaged.

[0009] In the information processing device, a search process that is performed by the processing device includes a first step of calculating, for each of the features included in the original data, a relative frequency distribution in the original data.

[0010] The search process includes a second step of setting a plurality of time windows for clipping data of a partial period of the original data in such a manner that a sum of periods of all the time windows is shorter than the predetermined period.

[0011] The search process includes a third step of clipping data from the original data according to the time windows.

[0012] The search process includes a fourth step of calculating, for each of the features, the relative frequency distribution in extracted data obtained by combining all the data clipped according to the time windows.

[0013] The search process includes a fifth step of calculating an error between the relative frequency distribution in the original data and the relative frequency distribution in the extracted data.

[0014] In the search process, after the first step is performed, a trial from the second step to the

fifth step is repeatedly performed by changing settings of the time windows. The processing device is configured to perform the search process to extract the extracted data with the error equal to or less than a threshold value.

[0015] The processing device is configured to calculate the index value using the extracted data with the error equal to or less than the threshold value.

[0016] In one aspect of the information processing device, the processing device may be configured to perform clustering. The clustering may be machine learning that groups data of each interval obtained by dividing the original data into intervals of a certain period into a predetermined number of clusters.

[0017] The processing device may be configured to set, in the second step, the time windows in such a manner that a difference between a proportion of each of the clusters in the extracted data and a proportion of each of the clusters in the entire original data is equal to or less than a threshold value.

[0018] The above information processing device can calculate the index value using the extracted data, namely the data whose data volume is smaller than that of the original data, with the same accuracy as in the case where the original data is used. Therefore, this information processing device can reduce the data volume while maintaining the accuracy, and can calculate the index value in a shorter time than in the case where the original data is used.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0020] FIG. 1 is a schematic diagram illustrating a relationship between a data center, a vehicle, and an information processing terminal, which is an embodiment of an information processing device;

[0021] FIG. 2 is a graph showing the original data, the upper figure showing a change in operating state of the engine-disconnecting clutch, the middle figure showing the relative rotational speed of the engine-disconnecting clutch, and the lower figure showing the transition of the inertia of the engine;

[0022] FIG. 3 is a flowchart illustrating a flow of processing performed by the processing device of the data center;

[0023] FIG. 4 is the graph which showed an example which clustered the original data using two features;

[0024] FIG. 5A shows an example of the relative frequency distribution in the original data, and it shows an example of the relative frequency distribution for the number of engagements of the engine-disconnecting clutch;

[0025] FIG. 5B shows an example of the relative frequency distribution in the original data, and it shows an example of the relative frequency distribution for the relative rotational speed of the engine-disconnecting clutch; and

[0026] FIG. 5C shows an example of the relative frequency distribution in the original data. It shows an example of the relative frequency distribution for the inertia of the engine.

DETAILED DESCRIPTION OF EMBODIMENTS

Configuration of Information Processing System

[0027] Hereinafter, a data center 500, which is an embodiment of an information processing device, will be described referring to FIG. 1 to FIG. 5C.

[0028] FIG. 1 shows a configuration of an information processing system including a data center

500. As shown in FIG. 1, the data center **500** communicates with the vehicle **10** via a communication network **400**. The data center **500** also communicates with the information processing terminal **600** via the communication network **400**. The data center **500** communicates with the plurality of vehicles **10** and the plurality of information processing terminals **600** via the communication network **400**.

Configuration of the Data Center **500**

[0029] As illustrated in FIG. 1, the data center **500** includes a processing device **510**. The data center **500** includes a storage device **520** and a communication device **530**. The processing device **510** includes a CPU that performs processing in accordance with a program, and a ROM in which the program is stored. The storage device **520** stores a large amount of data. The communication device **530** is implemented as hardware such as a network adapter, various communication software, or a combination thereof. The communication device **530** realizes wired or wireless communication via the communication network **400**.

Configuration of the Vehicle **10**

[0030] The data center **500** may be configured using a plurality of computers. For example, the data center **500** may be configured by a plurality of server apparatuses.

[0031] Each of the plurality of vehicles **10** includes a communication device **90**. The communication devices **90** are implemented as hardware such as a network adapter, various types of communication software, or a combination thereof. The communication devices **90** are configured to realize wired or wireless communication via the communication network **400**.

[0032] Each vehicle **10** includes an engine **20**, a motor **30**, a torque converter **40**, and an automatic transmission **50**. The output shaft of the engine **20** and the output shaft of the motor **30** are connected to the torque converter **40**. Torque of the engine **20** and the motor **30** is transmitted to the torque converter **40**. The torque transmitted to the torque converter **40** is transmitted to the automatic transmission **50**. The vehicle **10** is driven by the automatic transmission **50** transmitting the torque of the engine **20** and the motor **30** to the drive wheels of the vehicle **10**. The torque converter **40** and the automatic transmission **50** are operated by the hydraulic pressure of the hydraulic oil.

[0033] Each vehicle **10** includes an engine-disconnecting clutch **60**. The engine-disconnecting clutch **60** functions as a clutch for disconnecting the engine **20** from the torque transmission path of the vehicle **10**. The engine-disconnecting clutch **60** includes an input shaft **61** to which the rotation of the engine **20** is input, and an output shaft **62** which outputs the rotation of the engine **20**. The input shaft **61** rotates integrally with the output shaft of the engine **20**. The output shaft **62** rotates integrally with the output shaft of the motor **30**. The torque of the engine **20** is transmitted to the torque converter **40** via the input shaft **61** and the output shaft **62**. The torque of the motor **30** is transmitted to the torque converter **40** via the output shaft **62**.

[0034] The engine-disconnecting clutch **60** includes an input engagement member **63** coupled to the input shaft **61** and an output engagement member **64** coupled to the output shaft **62**. The output engagement member **64** is configured to be engageable with the input engagement member **63**. When the output engagement member **64** engages with the input engagement member **63**, rotation is transmitted between the input shaft **61** and the output shaft **62**.

[0035] For example, when the input engagement member **63** and the output engagement member **64** are pressed against each other, a frictional force is generated between the input engagement member **63** and the output engagement member **64**. By adjusting the frictional force, the operating state of the engine-disconnecting clutch **60** is switched. The operation state of the engine-disconnecting clutch **60** is switched by, for example, adjusting the frictional force between the input engagement member **63** and the output engagement member **64** by the hydraulic pressure of the hydraulic oil. The operating state of the engine-disconnecting clutch **60** includes a first engaged state, a second engaged state, and a disengaged state.

[0036] The first engaged state is a state in which rotation is transmitted between the input

engagement member **63** and the output engagement member **64** without the output engagement member **64** slipping with respect to the input engagement member **63**. In the first engaged state, the output engagement member **64** rotates integrally with the input engagement member **63**. In the first engaged state, a slight relative rotation may be allowed between the output engagement member **64** and the input engagement member **63**.

[0037] The second engaged state is a state in which rotation is transmitted between the input engagement member **63** and the output engagement member **64** while the output engagement member **64** slips with respect to the input engagement member **63**. In the second engaged state, the rotational speed of the output engagement member **64** is different from the rotational speed of the input engagement member **63**.

[0038] The disconnected state is a state in which no rotation is transmitted between the input engagement member **63** and the output engagement member **64**. In the disconnected state, the input engagement member **63** is not engaged with the output engagement member **64**, so that the transmission of torque between the engine **20** and the motor **30** is interrupted.

[0039] The vehicle **10** includes a control device **70**. The control device **70** controls the engine **20** and the motor **30**. The control device **70** controls the output torque of the engine **20** by controlling the throttle actuator, the fuel injection device, the ignition device, and the like of the engine **20**. The control device **70** controls the output torque of the motor **30** by controlling an inverter circuit provided between the motor **30** and the battery of the vehicle **10**.

[0040] The vehicle **10** includes a hydraulic control circuit **80**. The hydraulic control circuit **80** can change the hydraulic pressure of the hydraulic oil supplied to the engine-disconnecting clutch **60**. The control device **70** controls the hydraulic control circuit **80**. The control device **70** switches the operating state of the engine-disconnecting clutch **60** by changing the hydraulic pressure of the hydraulic oil supplied to the engine-disconnecting clutch **60**. The hydraulic control circuit **80** may further be capable of changing the hydraulic pressure of the hydraulic oil supplied to the torque converter **40** and the automatic transmission **50**. The control device **70** controls the torque converter **40** and the automatic transmission **50** by changing the hydraulic pressure of the hydraulic oil supplied to the torque converter **40** and the automatic transmission **50**.

[0041] The control device **70** is equipped with various sensors that collect information of each unit of the vehicle **10**.

[0042] In each vehicle **10**, travel data is collected from the various sensors. The traveling data is transmitted from each vehicle **10** to the data center **500** by the communication device **90**. For example, travel data including the travel distance, the position information, and the vehicle speed of each vehicle **10** is transmitted from each vehicle **10** to the data center **500**. Identification information for identifying the respective vehicles **10** is also transmitted from the respective vehicles **10** to the data center **500** together with the traveling data.

[0043] The data center **500** stores the traveling data together with the received identification information in the storage device **520**. In this way, traveling data of the plurality of vehicles **10** is accumulated in the storage device **520** of the data center **500**.

Configuration of the Information Processing Terminal **600**

[0044] The information processing terminal **600** includes a processing device **610**, a storage device **620**, and a communication device **630**. The processing device **610** includes a CPU that performs processing in accordance with a program, and a ROM in which the program is stored. The storage device **620** stores data. The communication device **630** is implemented as hardware such as a network adapter, various communication software, or a combination thereof. The communication device **630** realizes wired or wireless communication via the communication network **400**. The information processing terminal **600** is, for example, a personal computer.

Analysis of Travel Data of the Vehicle **10**

[0045] The information processing terminal **600** is used to analyze travel data. When analyzing the traveling data, an instruction for performing the analysis is transmitted from the information

processing terminal **600** to the data center **500**. The processing device **510** of the data center **500** that has received the instruction performs analysis using a part of travel data among the enormous travel data stored in the storage device **520** of the data center **500**. The travel data to be used is selected from the enormous amount of travel data stored in the storage device **520** in accordance with the purpose of analysis.

[0046] For example, the processing device **510** calculates a load applied to a specific component of the specific vehicle **10** based on travel data of the specific vehicle **10**. The processing device **510** estimates the damage accumulated in the component based on the calculated load. For example, the processing device **510** calculates an index value indicating the magnitude of the damage accumulated in the engine-disconnecting clutch **60** of the specific vehicle **10** based on the traveling data of the specific vehicle **10**. The processing device **510** of the data center **500** outputs the calculated result by transmitting the calculated result to the information processing terminal **600**. The information processing terminal **600** that has received the result displays the received result.

[0047] In order to perform such an analysis, the processing device **510** analyzes a large amount of travel data collected over a long period of time. Since the processing device **510** needs to perform an enormous amount of computation, it takes a long time to analyze.

[0048] Therefore, it is conceivable to extract the extracted data that captures the features of the entire original data from a large amount of travel data that is the original data. If such extracted data can be extracted, the processing device **510** can perform analysis in a shorter time by using the extracted data. For example, in the case of estimating the damage of the engine-disconnecting clutch **60** when the vehicle travels for 100,000 hours, the processing device **510** estimates the damage using the extracted data for 20,000 hours extracted from the original data for 100,000 hours. Then, the processing device **510** multiplies the index value calculated from the extracted data for 20,000 hours by 5 to calculate the index value of the damage of the engine-disconnecting clutch **60** when the vehicle travels for 100,000 hours.

[0049] FIG. 2 illustrates an example of original data. The original data is travel data for 100,000 hours in one vehicle **10**. FIG. 2 shows a part of original data which is travel data for 100,000 hours. The original data illustrated in FIG. 2 includes, as the feature, data of the number of engagements, the relative rotational speed, and the inertia of the engine-disconnecting clutch **60** to be subjected to calculation of the index value of damage.

[0050] The upper view of FIG. 2 shows a change in operating state of the engine-disconnecting clutch **60**. In the upper view of FIG. 2, a state where the operating state of the engine-disconnecting clutch **60** is the open state is indicated by “0”, and a state where the operating state of the engine-disconnecting clutch **60** is the first engaged state or the second engaged state is indicated by “1”. The number of engagements of the engine-disconnecting clutch **60** as the feature is the number of times that the operating state of the engine-disconnecting clutch **60** is in the first engaged state or the second engaged state in a predetermined period of time. The control device **70** calculates the number of engagements from the change in operating state of the engine-disconnecting clutch **60**. Data of the operating state of the engine-disconnecting clutch **60** may be transmitted from the vehicle **10** to the data center **500**, and the number of engagements may be calculated by the data center **500**.

[0051] FIG. 2 shows a change in relative rotational speed. The relative rotational speed as the feature is the relative rotational speed between the input shaft **61** and the output shaft **62** of the engine disconnecting clutch **60** when the engine-disconnecting clutch **60** is engaged. The expression “when the engine-disconnecting clutch **60** is engaged” means when the operating state of the engine-disconnecting clutch **60** changes from the disconnected state to the first engaged state or the second engaged state. The control device **70** acquires the relative rotational speed from the rotational speed of the output shaft of the engine **20** and the rotational speed of the output shaft of the motor **30**. From the vehicle **10**, data of the number of revolutions of the output shaft of the engine **20** and the number of revolutions of the output shaft of the motor **30** may be transmitted to

the data center **500**, and the relative rotational speed may be calculated in the data center **500**.

[0052] The number of engagements and the relative rotational speed are correlated with the damage of the engine-disconnecting clutch **60** of the vehicle **10**. The processing device **510** of the data center **500** estimates the damage of the engine-disconnecting clutch **60** from the travel data including the number of engagements and the relative rotational speed as the feature.

[0053] The lower figure of FIG. **2** shows the change in inertia for 100,000 hours. The inertia as the feature is the inertia of the engine **20** that outputs the rotational force to the engine-disconnecting clutch **60**. The control device **70** calculates inertia from the number of revolutions of the output shaft of the engine **20** when the operating state of the engine-disconnecting clutch **60** changes from the open state to the first engaged state or the second engaged state. From the vehicle **10**, data of the rotational speed of the output shaft of the engine **20** when the operating state of the engine-disconnecting clutch **60** becomes the first engaged state or the second engaged state may be transmitted to the data center **500**, and inertia may be calculated in the data center **500**.

[0054] The temperature of the engine **20** may be further used to calculate the inertia of the engine **20**. When the temperature of the engine **20** is low, the viscosity of the engine oil increases, and the inertia of the engine **20** increases. The control device **70** calculates the inertia larger as the temperature of the engine **20** is lower.

[0055] The inertia of the engine **20** correlates with the damage of the engine-disconnecting clutch **60** of the vehicle **10**. The processing device **510** of the data center **500** estimates the damage of the engine-disconnecting clutch **60** from the travel data including inertia in addition to the number of engagements and the relative rotational speed as the feature.

[0056] The extracted data is created by clipping the data from the original data by a plurality of time windows. In FIG. **2**, as an example of a part of the plurality of time windows, three time windows of the first time window **W_1**, the second time window **W_2**, and the third time window **W_3** are indicated by broken lines. The beginning and end of each time window are set such that the respective time windows do not overlap. In this example, the traveling data for 20,000 hours is clipped as the extracted data. Therefore, the start and end periods of each time window are set so that the total length of the time periods of all the time windows is 20,000 hours.

[0057] The data center **500** searches for the setting of the start time and the end time of each time window indicating the clipping pattern for extracting the extracted data that captures the features of the entire original data.

[0058] The data center **500** extracts the extracted data from the original data by using the clipping pattern found by the search. The data center **500** performs analysis using the extracted data.

Clipping Pattern Search Process

[0059] FIG. **3** is a flowchart illustrating a flow of a series of processes related to the clipping pattern search process. This series of processing is performed by the processing device **510** of the data center **500**.

[0060] As illustrated in FIG. **3**, the processing device **510** acquires the original data in the processing of **S100**. The original data is a part of the travel data selected for the purpose of analysis from the enormous travel data stored in the storage device **520** of the data center **500**. For example, the original data for calculating the index value indicating the magnitude of the damage accumulated in the engine-disconnecting clutch **60** of one vehicle **10** is travel data for a predetermined period of the target vehicle **10** selected from the huge travel data of the plurality of vehicles **10**. For example, in the case of estimating the damage of the engine-disconnecting clutch **60** when traveling for 100,000 hours, the original data is traveling data over a predetermined period of the target vehicle **10**.

[0061] In **S110** process, the processing device **510** labels the original data by clustering. Specifically, the processing device **510** divides the original data at regular intervals. The length of the period for separating the original data is, for example, several minutes. Then, the processing device **510** performs clustering which is machine learning for classifying the data of each interval

into a predetermined number of clusters. For example, k-means method is used as the algorithm of clustering. k-means method is a clustering algorithm for classifying data into a predetermined number of clusters. The clustering algorithm is not limited to k-means method.

[0062] The original data includes travel data collected under different environments, such as travel data when traveling in an urban area, travel data when traveling in a suburban area, and travel data when traveling on an expressway. By performing clustering, the travel data included in the original data can be classified into clusters of travel data having similar characteristics. The number of clusters to be classified is arbitrarily set according to the contents of the analysis.

[0063] FIG. 4 is a graph illustrating an exemplary clustering of original data into four clusters by a k-means method using two features included in the original data as explanatory variables. For example, the two features are the number of engagements and the relative rotational speed. In FIG. 4, each piece of data in each interval partitioned from the original data is indicated by a single point. When performing clustering, the processing device 510 uses a representative value of an explanatory variable in the data of each interval. For example, the processing device 510 sets the average value of the features in the data of each interval as a representative value. The processing device 510 may use, as the representative value, the moving average value of the features in a plurality of consecutive intervals in time series.

[0064] In FIG. 4, these points are shown in a two-dimensional space with the first feature FV_a and the second feature FV_b as coordinate axes. FIG. 4 is an example in which original data is clustered in four clusters of the first cluster M_1, the second cluster M_2, the third cluster M_3, and the fourth cluster M_4. In FIG. 4, the boundaries of the four clusters are indicated by solid lines. In FIG. 4, the center of gravity of each cluster is indicated by an open triangle. The center of gravity cgM_1 is the center of gravity of the first cluster M_1. The center of gravity cgM_2 is the center of gravity of the second cluster M_2. The center of gravity cgM_3 is the center of gravity of the third cluster M_3. The center of gravity cgM_4 is the center of gravity of the fourth cluster M_4.

[0065] Although FIG. 4 shows two examples of explanatory variables, the number of explanatory variables is not limited to two. For example, when the original data includes three features, the processing device 510 may perform clustering using these three features as explanatory variables. In this case, the processing device 510 clusters the original data in the three-dimensional space.

[0066] The processing device 510 assigns a label indicating the result of the clustering in this way to the original data. Specifically, each data indicated by a point in the coordinate space is given a label for identifying a cluster in which the data is classified. In this way, the processing device 510 creates the original data to which the label is attached.

[0067] Next, the processing device 510 calculates the relative frequency distribution of the original-data in the processing of S120. When a plurality of features is included in the original data, the processing device 510 calculates a relative frequency distribution in the original data for each feature.

[0068] The frequency distribution classifies data into a plurality of classes, and represents a frequency distribution that is the number of data of each class. The relative frequency indicates how much the frequency of the class accounts for the sum of the total frequencies.

[0069] FIG. 5A shows the relative frequency distribution for the number of engagements in the original data shown in FIG. 2. In this relative frequency distribution, the rank of the number of engagements in the original data is divided into m ranks from 1 to m, and the relative frequency distribution is shown.

[0070] FIG. 5B shows the relative frequency distribution for the relative rotational speed in the original-data shown in FIG. 2. In this relative frequency distribution, the relative frequency distribution is shown by dividing the class of the relative rotational speed in the original data into m classes from 1 to m.

[0071] FIG. 5C shows the relative frequency distribution for inertia in the original data shown in

FIG. 2. In this relative frequency distribution, the class of inertia in the original data is divided into m classes from 1 to m to show the relative frequency distribution.

[0072] In S120 process, the processing device 510 calculates the relative frequency distribution for the respective feature values included in the original data. The number of classes in the relative frequency distribution of each feature is the same.

[0073] Next, in S125 process, the processing device 510 sets a plurality of time windows in order to extract the extracted data from the original data.

[0074] In FIG. 2, three time windows W_1 to W_3 of the first time window W_1, the second time window W_2, and the third time window W_3 are shown as examples of a plurality of time windows. In the example shown in FIG. 2, the time periods of each time window are all equal. As illustrated in FIG. 2, the data clipped by each clipping window is data of a feature in the same period.

[0075] In S125 process, the processing device 510 randomly sets a plurality of time windows such that the total time period of all time windows is shorter than a predetermined period, which is the total time period of the original data. As will be described later, the processing device 510 combines all the data clipped by the plurality of time windows set here to generate extracted data. The total time period of all the time windows is a value for determining the capacity of the extracted data. Therefore, a period in which all the time windows are summed is set in advance.

[0076] For example, the processing device 510 randomly sets the number of time windows, the start of each time window, and the end of each time window each time S125 process is performed. At this time, the processing device 510 sets each time window so that each time window does not overlap. The processing device 510 thus randomly sets the plurality of time windows such that the total period of all time windows is a preset period. In S125 process, the processing device 510 may set a plurality of time windows by fixing the time periods of the time windows to be constant, as illustrated in FIG. 2. In S125 process, the processing device 510 may fix the plurality of time windows to a fixed number and set the plurality of time windows.

[0077] In addition to the above-described requirements, when setting a plurality of time windows through S125, the processing device 510 sets a plurality of time windows such that a difference between a proportion of each cluster in the extracted data and a proportion of each cluster in the entire original data is equal to or less than a threshold value.

[0078] In this way, by setting a plurality of time windows through S125 process, a clipping pattern in which data is clipped from the original data is determined. When the processing device 510 determines the clipping pattern in this way, the processing proceeds to S130.

[0079] In S130 process, the processing device 510 clips data from the original data in the determined clipping pattern. That is, in S130 process, the processing device 510 clips data from the original data by a plurality of set time windows. Then, the processing device 510 combines all the data clipped by the plurality of time windows to create extracted data.

[0080] In the process of the following S140, the processing device 510 calculates the relative frequency distribution of the extracted data. The processing device 510 calculates the relative frequency distribution of the extracted data in the same manner as the method of calculating the relative frequency distribution in S120. In other words, in S140 process, the processing device 510 calculates the relative frequency distribution of the extracted data for each feature value. At this time, the processing device 510 sets the number of grades in the relative frequency distribution of the respective features to be the same as the relative frequency distribution in S120.

[0081] Next, in S145 process, the processing device 510 calculates an error between the relative frequency distribution in the original data and the relative frequency distribution in the extracted data. For example, the processing device 510 calculates a mean absolute error MAE (Mean Absolute Error). The mean absolute error MAE is expressed by the following equation.

[00001]
$$MAE = \frac{1}{n} \cdot \sum_{i=1}^n \left| \frac{1}{m} \cdot \sum_{j=1}^m Y_{nm} - y_{nm} \right|$$
 Mathematical formula 1

In the above equation, “n” is the number of features. “m” is the number of series in the relative frequency distribution. “Y” is the frequency of the corresponding feature in the original data in the corresponding class. “y” is the frequency of the corresponding feature in the extracted data in the corresponding class.

[0082] As shown in the above equation, the processing device **510** calculates, as an error, the sum of the errors of the frequencies in the respective classes for each feature between the relative frequency distribution in the entire original data and the relative frequency distribution in the extracted data.

[0083] After calculating the error, the processing device **510** advances the processing to **S150**. In **S150** process, the processing device **510** determines whether or not the calculated error is less than or equal to the threshold value. The threshold value is a value for determining whether or not the extracted data having the relative frequency distribution close to the relative frequency distribution in the original data is extracted by the set clipping pattern. The magnitude of the threshold value is set in advance so that it can be determined that extracted data having a relative frequency distribution close to the relative frequency distribution in the original data is extracted based on the error being equal to or smaller than the threshold value.

[0084] In **S150** process, when it is determined that the error is equal to or smaller than the threshold value (**S150: YES**), the processing device **510** advances the process to **S160**.

[0085] In **S160** process, the processing device **510** calculates the target index value using the extracted data generated in the process of the latest **S130**. Here, an index value indicating the magnitude of the damage accumulated in the engine-disconnecting clutch **60** is calculated. For example, the processing device **510** calculates the degree of damage as an index value indicating the magnitude of damage accumulated in the engine-disconnecting clutch **60**.

[0086] The degree of damage is an index value representing the ratio of damage accumulated, assuming that the damage of the engine-disconnecting clutch **60** gradually accumulates, assuming that the damage resulting in damage is “1”. Here, the damage applied to the engine-disconnecting clutch **60** during a certain period of time is calculated based on the data of the number of engagements. In addition to the number of engagements, the damage applied to the engine-disconnecting clutch **60** from at least one of the relative rotational speed and the inertia may be calculated as the feature. Then, the degree of damage that the engine-disconnecting clutch **60** is damaged is set to “1”, and the calculated ratio of damage is calculated as an index value. By repeating this process, the degree of damage, which is the ratio of accumulated damage to damage leading to damage, is calculated. When the degree of damage becomes “1”, the damage is caused, and the calculated degree of damage is a value from “0” to “1”.

[0087] Here, since the damage degree is calculated using the extracted data which is a part of the original data, the processing device **510** converts the calculated damage degree into a size corresponding to the original data, and calculates the damage degree as the index value. For example, when the original data is traveling data for 100,000 hours and the extracted data is traveling data for 20,000 hours, the calculated degree of damage is multiplied by 5 to obtain the degree of damage as the index value.

[0088] On the other hand, in **S150** process, when it is determined that the error is larger than the threshold value (**S150: NO**), the processing device **510** returns the process to **S125**. Then, the processing device **510** performs the search processing up to **S125** to **S145** again.

[0089] In this way, the processing device **510** repeatedly performs **S125** to **S145** search processing by changing the settings of the plurality of time windows, and extracts extracted data in which the error becomes equal to or less than the threshold value from the original data. Then, the processing device **510** calculates an index value using the extracted data. After calculating the index value, the processing device **510** advances the processing to **S170**.

[0090] In **S170** process, the processing device **510** determines whether or not the index value is equal to or greater than a predetermined value. The default value is a value for predicting that

damage is more likely to occur based on the fact that the index value is equal to or larger than the default value. For example, “0.9” can be set here, for example, as a default value in the degree of damage. In this case, it is possible to predict that the possibility of the damage is high based on the fact that the damage has reached 90% of the damage leading to the damage.

[0091] In **S170** process, when it is determined that the index value is equal to or greater than the predetermined value (**S170**: YES), the processing device **510** advances the process to **S180**. In **S180** process, the processing device **510** outputs an index value and a failure estimate. Specifically, the processing device **510** transmits the index value and the failure prediction to the information processing terminal **600** that has transmitted the instruction for requesting the analysis.

[0092] The failure prediction is, for example, a message indicating that the occurrence of a failure has been predicted. In this way, when the calculated index value is equal to or greater than the predetermined value, the processing device **510** notifies that the occurrence of the failure has been predicted. The failure prediction may be information of a lifetime until a failure occurs. For example, when the degree of damage calculated by using the extracted data extracted from the original data for 100,000 hours is the index value, the processing device **510** calculates the traveling time until the degree of damage reaches “1” and outputs the calculated traveling time as the information of the life. The information on the life may be converted into the traveling distance based on the traveling distance of 100,000 hours and output.

[0093] In **S170** process, when it is determined that the index value is less than the predetermined value (**S170**: NO), the processing device **510** advances the process to **S190**. In **S190** process, the processing device **510** outputs an index value. Specifically, the processing device **510** transmits the index value to the information processing terminal **600** that has transmitted the instruction for requesting the analysis.

Operation of this Embodiment

[0094] When **S180** or **S190** process is performed, the processing device **510** terminates the series of processes.

[0095] The data center **500**, which is the information processing device of the present embodiment, acquires original data collected and created over a predetermined period using a plurality of sensors mounted on the vehicle **10**, and calculates an index value indicating the magnitude of damage accumulated in the engine-disconnecting clutch **60**.

[0096] The data center **500** includes a processing device **510** that performs processing. The original data includes data of the number of engagements and the relative rotational speed of the engine-disconnecting clutch **60** as the feature. The original data further includes inertia data as a feature in addition to the number of engagements and the relative rotational speed. In the data center **500**, the search process performed by the processing device **510** includes a first step (**S120**) of calculating, for each feature, a relative frequency distribution in the original data for a plurality of features included in the original data. The searching process includes a second step (**S125**) of setting a plurality of time windows for clipping data of a part of the period of the original data such that the period of time of all the time windows is less than the predetermined period of time. The search process includes a third step (**S130**) of clipping data from the original data by a plurality of time windows. The search process includes a fourth step (**S140**) of calculating, for each feature, the relative frequency distribution in the extracted data obtained by combining all the data clipped by the plurality of time windows. The search process includes a fifth step (**S145**) of calculating an error between the relative frequency distribution in the original data and the relative frequency distribution in the extracted data. After performing the first step, the processing device **510** performs a search process in which a trial from the second step to the fifth step are repeatedly performed by changing the settings of a plurality of time windows. Then, the processing device **510** extracts the extracted data in which the error is equal to or less than the threshold value (**S150**: YES). The processing device **510** calculates an index value using the extracted data in which the error becomes equal to or smaller than the threshold value (**S160**).

[0097] According to the data center **500**, it is possible to obtain extracted data in which features of the entire original data including a plurality of features are captured. Therefore, the data center **500** can calculate the index value with the same accuracy as in the case of using the original data by using the extracted data having a smaller data amount than the original data.

Effect of this Embodiment

[0098] (1) According to the data center **500** that is the information processing device of the present embodiment, it is possible to achieve both reduction in the amount of data and calculation accuracy of the index value. [0099] (2) According to the data center **500** which is the information processing device of the present embodiment, it is possible to calculate the index value in a shorter time than in the case where the original data is used. [0100] (3) The processing device **510** performs clustering, which is machine learning for classifying the data of the sections obtained by dividing the original data into a predetermined number of clusters at regular intervals (**S110**). Then, in the second step (**S125**) of the search process, the processing device **510** sets a plurality of time windows such that the difference between the proportion of each cluster in the extracted data and the proportion of each cluster in the entire original data is equal to or less than the threshold value. [0101] A plurality of sections classified into the same cluster are intervals having similar characteristics. In the above-described search process, the setting output from the processing device **510** is a setting in which the difference between the proportion of the entire original data and each cluster is equal to or less than the threshold value, and the extracted data having the relative frequency distribution of each feature close to each other can be clipped.

[0102] Therefore, according to the search process performed by the data center **500**, it is possible to find a setting that can obtain extracted data closer to the characteristics of the entire original data.

[0103] (4) The processing device **510** terminates the search process when one piece of extracted data whose error becomes equal to or smaller than the threshold value can be extracted, and calculates an index value using the extracted data whose error becomes equal to or smaller than the threshold value. Therefore, the data center **500** can calculate an index value at a time point when one piece of extracted data whose error becomes equal to or smaller than the threshold value can be extracted, and output the result promptly. [0104] (5) When the calculated index value is equal to or greater than the predetermined value (**S170**: YES), the processing device **510** notifies that a failure has been predicted. Therefore, the data center **500** can notify the user that the occurrence of the failure has been predicted before the failure occurs. [0105] (6) The processing device **510** calculates a degree of damage as an index value. Therefore, the data center **500** can inform the user of how long the delay until the failure is reached.

Example of Change

[0106] The present embodiment can be modified to be implemented as follows. The present embodiment and modifications described below may be carried out in combination within a technically consistent range. [0107] Although the number of engagements of the engine-disconnecting clutch **60** to be subjected to calculation of the index value of damage, the relative rotational speed of the engine-disconnecting clutch **60**, and the inertia of the engine **20** have been exemplified as the feature, the index value may be calculated by including the data of the oil temperature of the hydraulic oil supplied to the engine-disconnecting clutch **60** as the feature. [0108] In the above embodiment, an example in which the information processing device is embodied as the data center **500** has been described. An example in which the index value is calculated in the data center **500** has been described. On the other hand, the information processing device described above may be embodied as the information processing terminal **600**. In this case, the calculation of the index value is performed by the processing device **610** of the information processing terminal **600**. The above-described information processing device may be embodied as the control device **70** of the vehicle **10**. In this case, the calculation of the index value can also be performed by the control device **70** of the vehicle **10**. [0109] In the above-described embodiment, an example has been described in which one piece of extracted data is extracted and an index value

is calculated. On the other hand, a final index value may be determined by extracting a plurality of pieces of extracted data and using a plurality of index values calculated using the respective pieces of extracted data. For example, the minimum value, the maximum value, the mode value, and the average value are set as final index values. Further, a plurality of index values may be output.

[0110] In the above-described embodiment, an example has been described in which, when the index value is equal to or larger than a predetermined value, it is notified that the occurrence of a failure has been predicted. This may be omitted. After the index value is calculated, only **S190** process may be performed, and only the index value may be outputted. [0111] Although the degree of damage is exemplified as an example of the index value to be calculated, the index value to be calculated is not limited to the degree of damage. [0112] The processing device **510** sets a plurality of time windows such that a difference between the proportion of each cluster in the original data and the proportion of each cluster in the extracted data is equal to or smaller than a threshold value. Without such a restriction, the processing device **510** may set a plurality of time windows. In such cases, the process **S110** clustering may be omitted. [0113] The method of determining the setting of the time window in the clipping pattern may not be random. The trial may be repeated by changing the setting of the time window in the clipping pattern according to a preset rule. [0114] The error calculated in **S145** process is not limited to the mean absolute error MAE. For example, the processing device **510** may calculate a mean square error as an error. The processing device **510** may calculate a root mean square error as an error. [0115] An example using extracted data obtained by combining all clipped data is shown. On the other hand, some of the clipped data may be combined to create extracted data.

Claims

1. An information processing device that acquires original data collected and created over a predetermined period using a plurality of sensors mounted on a vehicle and calculates an index value indicating a magnitude of damage accumulated in an engine-disconnecting clutch, the information processing device comprising a processing device configured to perform a process, wherein: the original data includes, as features, the number of engagements of the engine-disconnecting clutch and a relative rotational speed between an input shaft and an output shaft of the engine-disconnecting clutch when the engine-disconnecting clutch is engaged; the processing device is configured to perform a search process, the search process including a first step of calculating, for each of the features included in the original data, a relative frequency distribution in the original data, a second step of setting a plurality of time windows for clipping data of a partial period of the original data in such a manner that a sum of periods of all the time windows is shorter than the predetermined period, a third step of clipping data from the original data according to the time windows, a fourth step of calculating, for each of the features, the relative frequency distribution in extracted data obtained by combining all the data clipped according to the time windows, and a fifth step of calculating an error between the relative frequency distribution in the original data and the relative frequency distribution in the extracted data, and after the first step is performed, a trial from the second step to the fifth step being repeatedly performed by changing settings of the time windows, and the processing device performing the search process to extract the extracted data with the error equal to or less than a threshold value; and the processing device is configured to calculate the index value using the extracted data with the error equal to or less than the threshold value.
2. The information processing device according to claim 1, wherein the original data includes, as a feature, data on inertia of an engine that outputs a rotational force to the engine-disconnecting clutch.
3. The information processing device according to claim 1, wherein: the processing device is configured to perform clustering, the clustering being machine learning that groups data of each

interval obtained by dividing the original data into intervals of a certain period into a predetermined number of clusters; and the processing device is configured to set, in the second step, the time windows in such a manner that a difference between a proportion of each of the clusters in the extracted data and a proportion of each of the clusters in an entirety of the original data is equal to or less than a threshold value.

4. The information processing device according to claim 1, wherein the processing device is configured to end the search process when one piece of the extracted data with the error equal to or less than the threshold value is extracted, and calculate the index value using the piece of the extracted data with the error equal to or less than the threshold value.

5. The information processing device according to claim 1, wherein the processing device is configured to, when the calculated index value is equal to or greater than a predetermined value, notify that a failure is predicted to occur.
